

Adarsh Bharathwaj

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EDUCATION

University of Michigan

B.S.E. Computer Science and B.S. Mathematics (Honors), GPA: 3.86/4.0

Ann Arbor, MI

Aug. 2023 – May 2027

PUBLICATIONS

S. Kumar, **A. Bharathwaj**, D. Jurgens. “Cooperative Profiles Predict Multi-Agent LLM Team Performance in AI for Science Workflows.” *Conference on Language Modeling (COLM)*, 2026. arXiv:2604.20658

RELEVANT COURSEWORK

Machine Learning & Mathematics: Graduate Machine Learning, Graduate Probability Theory, Advanced Linear Algebra, Complex & Real Analysis, Computational Mathematics, Differential Equations

Systems & Theory: Data Structures & Algorithms, Operating Systems, Computer Organization, Theoretical CS, Database Management

EXPERIENCE

Software Engineering Intern & Co-Op

Jan. 2026 – Dec. 2026

JPMorgan Chase

Chicago, IL

- Built an automation pipeline on a four-person team converting discovery materials (meeting transcripts and SOPs) into AI-executable runbooks, cutting development time from one week to one day
- Independently implemented three agent-based pipelines for runbook conversion and MCP tool creation using multi-stage prompting, validating outputs against in-production documents with SME review
- Deployed the tool as a live internal web application through Jenkins CI/CD and authored user documentation plus a Claude skill that updates it automatically
- Raised runbook coverage from 50-60% to 95% through an LLM-as-a-judge feedback loop and cut generation time from 2-4 hours to 20-30 minutes

Generative AI and Natural Language Processing Researcher

May 2025 – Present

The Blablablab Research Group - University of Michigan

Ann Arbor, MI

- Proposed the research direction and sourced the Weakest-Link and O-Ring games; benchmarked 35 open-weight LLMs (0.6B–70B across 10 families) over six behavioral-economics games on Michigan’s HPC cluster with vLLM, producing 87,475 simulation-level observations
- Built the AI-for-Science evaluation: 7,310 LangGraph multi-agent simulations across 1,462 budget, visibility, and team-size conditions, scored by a dual frontier-model LLM-as-judge protocol (inter-judge $r = 0.88$ on quality)
- Established cooperative profile as a deployment screen via Bayesian hierarchical measurement-error models in NumPyro, replacing the full agent pipeline at 44x lower latency and 27x fewer tokens

PROJECTS

Low-Latency Market Data Replication Engine | C++20, UDP, Concurrency, Benchmarking

Aug. 2026 – Present

- Built a cross-platform C++20 UDP publisher-receiver for sequenced L2 options book updates with concurrent ACK-collector threads, gap detection, and atomic book commits
- Benchmarked receiver service-time, throughput, and UDP-payload efficiency across 6 batching configurations with fixed-seed replay and state-hash correctness checks, 5 runs per configuration

Agentic Framework for DeepSeek R1 API | Python, Multimodal Agents, Vector Memory, Open-Source

Jan. 2025 – Jun. 2025

- Built a multimodal framework over the DeepSeek R1 API with long-term memory, enabling multi-step planning and autonomous web navigation
- Benchmarked the framework against baseline R1, achieving 5%-10% higher scores on WebShop and WebArena

Brilliant Detroit | RAG, LangChain, Pinecone, Google APIs, Python

Feb. 2025 – Apr. 2025

- Built a retrieval-augmented generation pipeline over a nonprofit’s Drive, embedding thousands of documents across multiple formats into a Pinecone vector index with fuzzy tag search

MiniCopilot | PyTorch, GPT-2, Hugging Face, AWS Lambda, Docker, TypeScript

Sep. 2024 – Dec. 2024

- Trained a GPT-2 architecture code completion model from scratch on HuggingFace code datasets with a small team
- Deployed the model on AWS behind Lambda-invoked Docker containers, serving completions to a custom VSCode extension built in TypeScript

HONORS & AWARDS

- ICPC ECNA: T40 (2025)
- MHacks 2024 Winner
- James B. Angell Scholar
- Dean’s List (6x)

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, C, JavaScript/TypeScript

ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, vLLM, LangChain, LangGraph, RAG, LLM-as-a-judge evaluation, MCP, NumPyro, NumPy, Pandas

Systems/Tools: Git, Docker, AWS (Lambda, S3), Jenkins CI/CD, REST APIs, distributed systems, multithreading, Linux CLI, React, Next.js